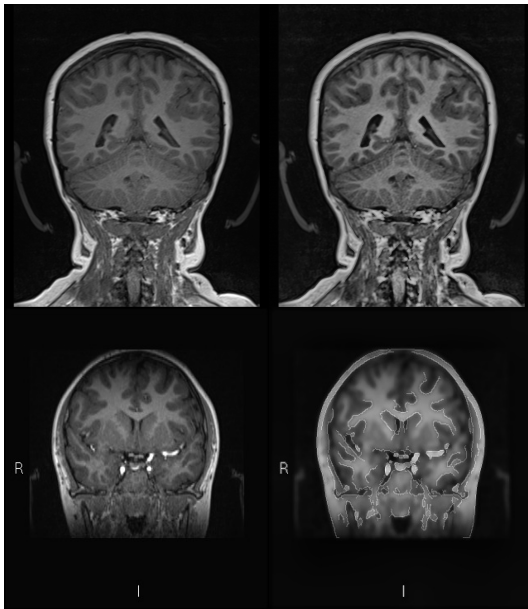




Specialcourse NRU

Holger Max Fløe Lyng

Investigating Methods for Polymicrogyria Classification

A replication of Guha et al. “Automated detection of polymicrogyria in pediatric patients using deep learning”

Dato: Juni 2026

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1 Introduction

Deep learning has transformed medical image analysis, achieving performance comparable to clinical experts in different tasks such as grading diabetic retinopathy, classification of skin lesions, and radiological classification (Litjens et al., 2017). Convolutional neural networks have been applied to automated detection of structural brain abnormalities from magnetic resonance imaging, with the promise of supporting radiologists in high-volume screening and reducing diagnostic delays in remote areas. However, this requires that reported performance metrics are grounded in sound methodology, free from data leakage, label noise, and low-level image confounds that can inflate accuracy without reflecting genuine pathology signal detection.

Polymicrogyria (PMG) is a malformation of cortical development characterised by excessive cortical folding and abnormal lamination (Severino et al., 2020). It is one of the most common malformations of cortical development, accounting for approximately 20% of all such malformations (Stutterd et al., 1993). PMG is frequently encountered in patients with drug-resistant epilepsy and conditions is commonly associated with neurodevelopmental delay, motor impairments, and cognitive deficits. PMG remains challenging to characterise reliably due to its heterogeneous radiological appearance, variable distribution of anatomical features, and the limited availability of curated neuroimaging datasets.

Deep learning attempts to classify PMG have predominantly been conducted on small, imbalanced cohorts and without proper patient-level data splitting (Guha et al., 2025), making it difficult to distinguish genuine classification performance from artifacts of data leakage, class imbalance, or low-level image difference introduced during acquisition. Furthermore, the use of models pretrained on natural image datasets — comprising everyday photographs of cars, birds, and household objects — introduces a substantial domain gap relative to clinical neuroimaging; ImageNet-pretrained features have been shown to transfer poorly beyond the lowest convolutional layers, offering little benefit over task-specific training (Raghu et al., 2019), which may further limit the sensitivity of these models to the subtle cortical morphology features relevant to PMG classification.

However, automated detection of PMG from MRI could inform clinical decision-making by reducing the diagnostic burden on specialist neuroradiologists and enabling earlier detection and identification of previously missed cases through the detection of subtle features. Fortunately, a publicly available benchmark dataset — the Pediatric Polymicrogyria MRI (PPMR) dataset published by Zhang et al. (2024) — has been released, and two independent studies have applied deep learning methods to PMG classification using this resource (Guha et al., 2025; Zhang et al., 2024).

This study undertakes an independent replication and methodological evaluation of Guha et al. (2025). We examine the assumptions underlying their experimental design, assess whether the reported near-perfect classification accuracy can be attributed to genuine PMG detection, and quantify the effect of applying corrective measures to the identified methodological issues. We identify three methodological errors in Guha et al.: (1) Label noise from including PMG-negative slices in the positive class, (2) a systematic class-correlated resolution difference that provides a shortcut signal, and (3) pre-split HC downsampling that produces artificially balanced evaluation sets and ignores patient-level integrity.

In addition, we evaluate the corrected pipeline on an independent clinical dataset of epilepsy patients, providing an assessment of generalisation beyond the PPMR benchmark.

2 Background

This chapter establishes the background needed to interpret the replication. Section 2.1 describes the pathology of polymicrogyria and why its imaging features are difficult to detect. Section 2.2 introduces T1-weighted MRI. Section 2.3 covers the convolutional architectures used here, ResNet-101 and DenseNet-201, together with transfer learning. Section 3.2 then introduces Guha et al. (2025) and the three methodological errors this project sets out to correct.

2.1 Polymicrogyria

Polymicrogyria (PMG) is a malformation of the developing brain characterised by abnormal cortical lamination and an excessive number of abnormally small gyri, resulting in an irregular folding pattern across all or part of the cerebral cortex (“Polymicrogyria”, n.d.; Stutterd et al., 1993). PMG arises during the post-migrational stage of cortical development, distinguishing it from other malformations of cortical development (MCDs) such as lissencephaly, which affects the earlier migration stage (Severino et al., 2020). Multiple forms of PMG have been identified. Unilateral PMG affects one hemisphere only, whereas bilateral PMG involves both hemispheres, and severity ranges from mildly focal to extensive bilateral involvement. PMG accounts for approximately 20% of all MCDs, making it one of the most common brain malformations (Stutterd et al., 1993).

Despite its prevalence among MCDs, PMG remains challenging to diagnose reliably. MRI is the only imaging modality that provides sufficient soft-tissue contrast and spatial resolution to identify the fine cortical folds characteristic of PMG; computed tomography (CT) is inadequate for this task (Severino et al., 2020). The radiological appearance of PMG is heterogeneous, e.g. features include an irregular or bumpy cortical surface, a stippled gray-white matter junction, and apparent cortical thickening. In some cases of PMG, adjacent microgyri fuse their overlying molecular layers, producing a deceptively smooth outer surface that can be mistaken for other MCDs. MRI appearance also changes with myelination, making diagnosis difficult in infants during the first year of life, when low grey-white matter contrast may obscure the condition’s subtle cortical irregularities. Inter-rater variability among specialist neuro-radiologists is well documented, underscoring both the difficulty of the diagnostic task and the potential value of automated detection methods.

2.2 MRI

Magnetic Resonance Imaging (MRI) is the principal structural neuroimaging modality across a wide range of neurological conditions, including epilepsy, brain tumors, white matter diseases, and cortical malformations. Its clinical dominance in neurology stems from its capacity to provide high soft-tissue contrast without ionising radiation. T1-weighted sequences in particular offer excellent differentiation between gray and white matter, enabling delineation of cortical thickness, sulcal morphology, and the gray-white matter junction architecture that is central to identifying MCDs such as PMG.

A key challenge in applying deep learning to retrospective or multi-site neuroimaging data is scanner variability. Differences in field strength, scanner vendor, pulse sequence, and field of view produce systematic differences in signal intensity, spatial resolution, and image contrast that are unrelated to the pathology of interest. In datasets where patient groups were not acquired under matched conditions, these low-level statistical differences can act as confounds, allowing classifiers to exploit acquisition-related

signals rather than anatomical features. For this project this is a relevant risk, since the PPMR dataset was acquired on two scanners at different field strengths.

2.3 Deep learning for Medical Image Classification

Convolutional neural networks (CNNs) learn a hierarchical image representation — edges and textures in early layers, semantic structure in deeper ones – efficiently, through local connectivity and weight sharing (Litjens et al., 2017). The two architectures evaluated here, ResNet-101 and DenseNet-201, are pretrained on ImageNet and described below.

2.3.1 ResNet-101

He et al. (2016) introduced the residual block, in which a layer learns the residual mapping $F(x) = H(x) - x$ so that its output is $H(x) = F(x) + x$, as shown in Figure 1. The skip connection means the block only needs to learn the deviation of the residual, which eases the optimisation of very deep networks and lets gradient flow directly to earlier layers during backpropagation, avoiding the so-called exploding- and vanishing-gradient problem. ResNet-101 comprises of 101 weighted layers organised in four residual stages, achieving state-of-the-art performance on ImageNet ILSVRC 2015 and has become a widely adopted backbone for medical image classification tasks.

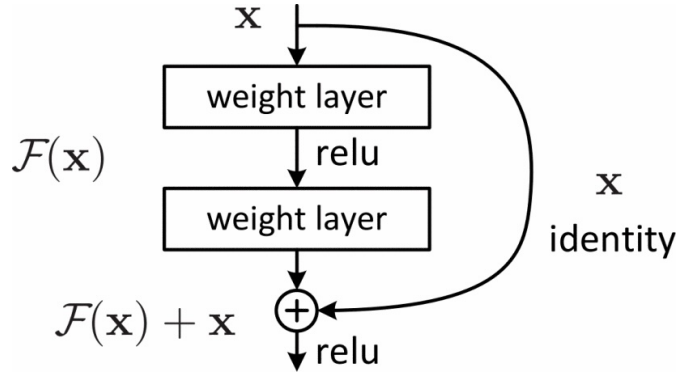


Figure 1: Residual learning: A building block (He et al., 2016)

2.3.2 DenseNet-201

In a dense block each layer receives the concatenated feature maps of all preceding layers, yielding $L(L + 1)/2$ connections for L layers (Huang et al., 2017). In Figure 2, an individual dense block is shown (top), and a full DenseNet is shown (bottom). This dense connectivity encourages feature reuse, strengthens gradient flow, and reduces the number of parameters compared with comparably deep networks. DenseNet-201 is the 201-layer variant built from four dense blocks. It was the best-performing PPMR model reported by Guha et al. (2025).

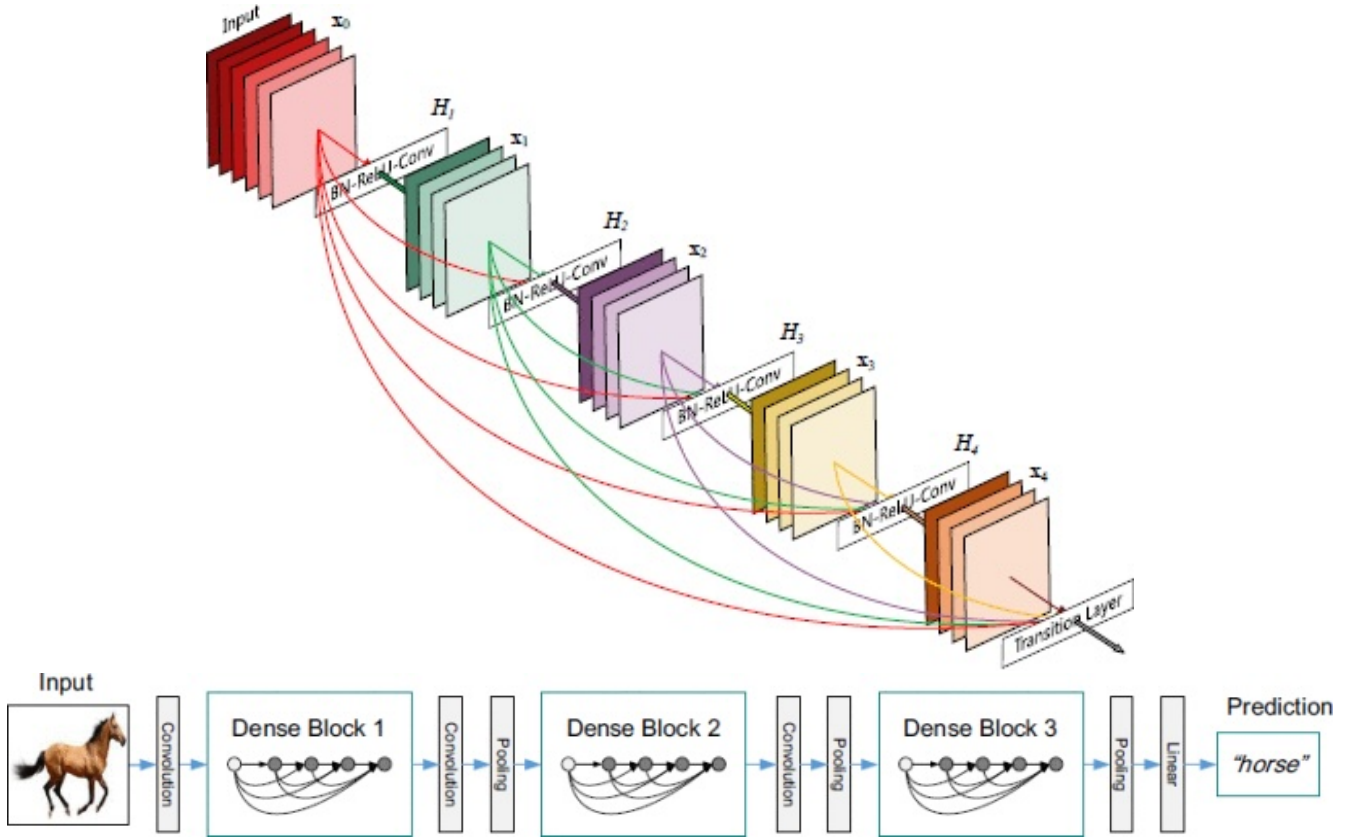


Figure 2: Top: A 5-layer dense block with a growth rate of $k=4$. Each layer takes all preceding feature-maps as input (Huang et al., 2017). Bottom: A deep DenseNet with three dense blocks. The layers between two adjacent blocks are referred to as transition layers and change feature-map sizes via convolution and pooling (Huang et al., 2017).

2.3.3 Transfer learning

When training a network, there are two options: start from scratch or use a pretrained backbone. Training from scratch requires large annotated datasets, which are not available for PMG data. Transfer learning sidesteps the availability issue by initialising from a pretrained model’s weights, in this case trained on ImageNet data (approximately 1.2 million images across 1000 classes) and fine-tuning on the target task (Raghu et al., 2019). A pretrained network already knows how to detect general features such as edges and textures, and these transfer well to MRI; only its object-specific high-level features do not. This is why fine-tuning succeeds on small medical datasets, the network can reuse the weights and only learns to map those features to the PMG-versus-control decision. But this reliance on low-level features is also its weakness, as the model can pick up on texture differences caused by how the images were acquired or preprocessed rather than pathology; this foreshadows the shortcut learning.

3 Methods

3.1 Data

3.1.1 PPMR

This project uses the publicly available PPMR dataset introduced by Zhang et al. (2024) and subsequently used by Guha et al. (2025). The dataset consists of coronal 2D gradient-echo T1-weighted MRI slices from 23 pediatric epilepsy patients with confirmed PMG and three age and gender-matched HCs (HC) per patient, yielding a 3:1 control-to-case patient ratio.

A critical property of the dataset is that not all MRI slices from PMG patients contain visible malformation. Each slice was individually labelled by a pediatric neuroradiologist as PMG-positive (label 1), PMG-negative (label 2), or uncertain (label 3) (Zhang et al., 2024). As shown in Figure 3, only approximately 49% of all PMG-patient slices are labelled as PMG-positive; approximately 30% are PMG-negative, and approximately 19% are labeled uncertain.

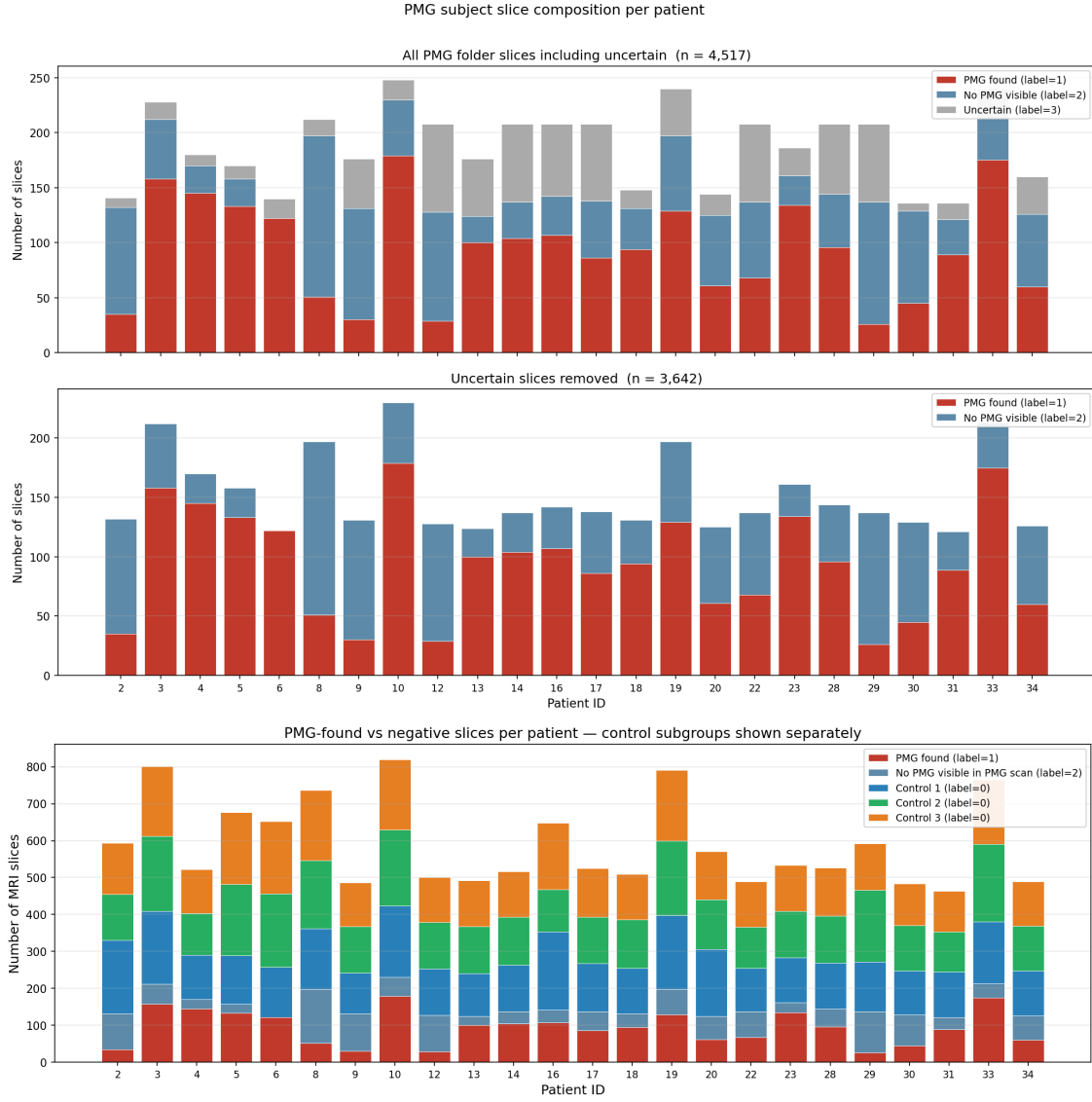


Figure 3: Top: Number of 2D slices per PMG patient labelled as PMG-positive (red), PMG-negative (blue), and uncertain (grey), based on the Kaggle release of the PPMR dataset Zhang et al., 2024. Middle: The same distribution excluding uncertain slices, replicating the figure published in the original dataset paper Zhang et al., 2024. Bottom: Total distribution of slices across control and PMG cohorts Zhang et al., 2024.

Aggregating across all labels, the dataset contains 4517 2D slices attributed to PMG patients and 10539 slices attributed to HCs. In figure 4 are the full statistics for the dataset.

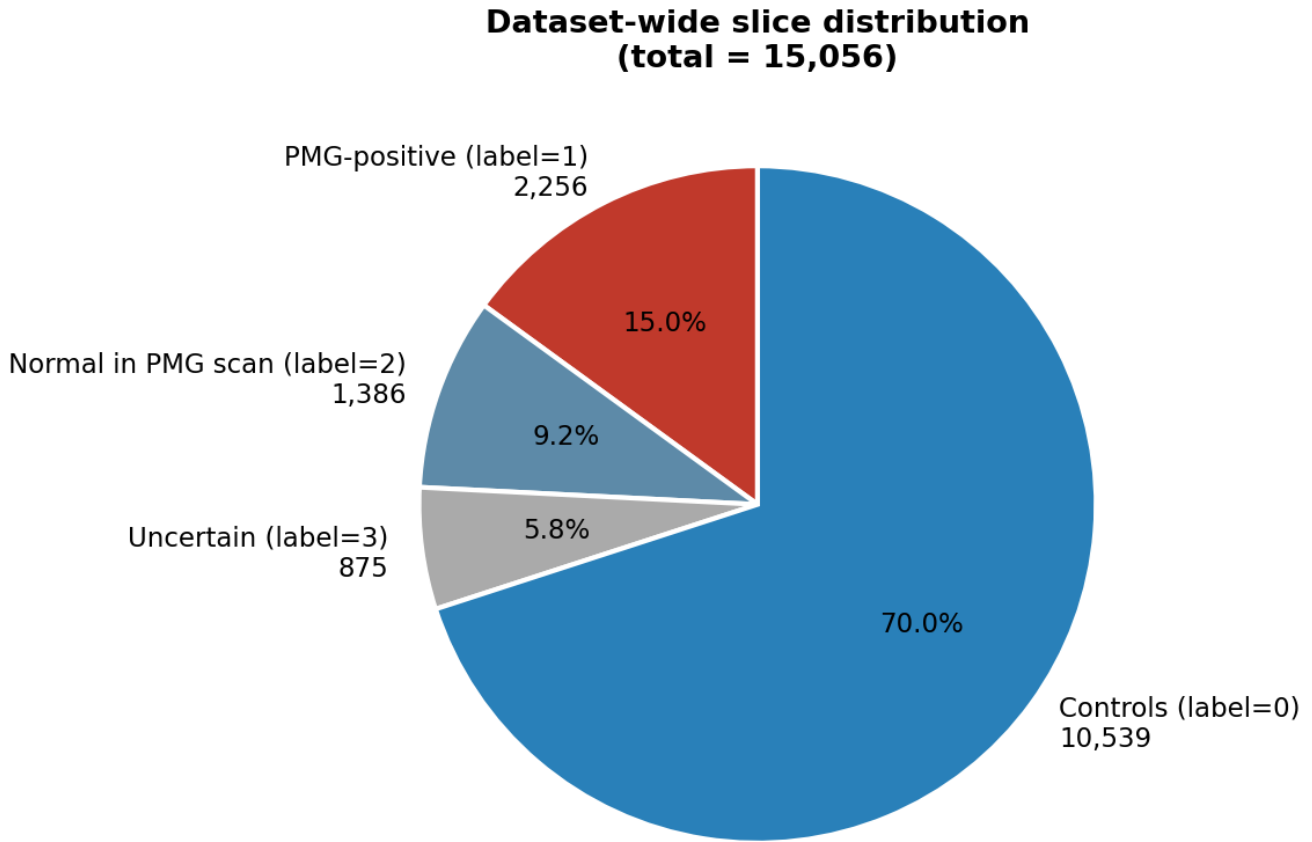


Figure 4: The full label distribution of the dataset published on Kaggle (Zhang et al., 2024). Guha et al. used this data distribution

As defined in the dataset release paper, the dataset was labeled by a pediatric neuroradiologist with 12 years of experience reading children’s brain MRIs (Zhang et al., 2024). The patient was included in the study if PMG was found. The MRIs were acquired using the highest resolution sequence for displaying cortical details, the coronal 3D gradient echo T1 weighted sequence, these volumes were exported as JPEG images from the PACS system. The imaging parameters used for acquisition of the coronal T1 weighted images are shown in Table 1.

Parameter	3 T Siemens Skyra	1.5 T GE Cigna
Sequence	Coronal 3D GRE T1	Coronal 3D GRE T1
Repetition time (TR)	2200 ms	10.44 ms
Inversion time (TI)	1030 ms	450 ms
Echo time (TE)	2.63 ms	4.3 ms
Matrix	320 × 260	512 × 512
Slice thickness	1.2 mm	1.2 mm
Field of view (FOV)	20 × 23 cm	22 × 27 cm

Table 1: Acquisition parameters — PPMR dataset Zhang et al., 2024.

3.1.2 NRU data

To evaluate the preprocessing method and DL models used for the PMG classification task we introduce an in-house dataset. The dataset includes 183 T1-weighted MRI scans from 162 unique patients evaluated for epilepsy (77 females; mean $\pm$ standard deviation (SD) age at acquisition: 25.8 ± 17.4 years). Of these, 110 scans (49 females; 24.7 ± 15.9 years) were reported to contain radiological evidence of PMG based on clinical radiology reports. The non-PMG group (37 females; 27.5 ± 19.6 years) consisted of scans in which PMG-related terminology appeared in the radiology report, but a subsequent double-rater confirmed the absence of PMG. Of these, 56 cases had other abnormalities and 17 were reported as having a normal MRI. MRI scans were acquired in routine clinical practice and retrospectively collected from the Eastern Region of Denmark, encompassing multiple scanner protocols, two magnetic field strengths (B), and three vendors. The acquisition parameters are summarised in Table 2.

Parameter	Value
Scanner	Siemens Verio, 3 T
Sequence	MPRAGE (T1-weighted)
Repetition time (TR)	1900 ms
Echo time (TE)	2.23 ms
Inversion time (TI)	900 ms
Flip angle	9°
Voxel spacing	$1 \times 1 \times 1 \text{ mm}^3$

Table 2: Acquisition parameters — Hvidovre Hospital dataset.

The NRU dataset serves as an independent baseline against which the PPMR-trained pipeline is compared. The scans were acquired as 3D volumes and sliced into 2D coronal slices for the present 2D models. No slice-level annotations are available, only volume-level, so it follows the same convention as Guha et al. (2025), every slices from a PMG patient is treated as PMG-positive. The volumetric form retained deliberately, with the longer-term aim of training a 3D model directly on the volumes rather than on 2D slices.

3.2 Guha et al.

Guha et al. (2025) fine-tuned ImageNet-pretrained CNNs on the PPMR dataset to classify paediatric MRI slices as PMG or HC. Their best model, DenseNet-201, reported a near-perfect accuracy of 0.982 ± 0.034 , precision of 0.995 ± 0.005 , and recall of 0.947 ± 0.110 . We identified three methodological errors. (1) Label noise, all 4517 slices from PMG patients were treated as positive, even though only 49% ($n \approx 2256$) is annotated PMG-positive (label=1). (2) A class correlated resolution confound, enhanced by the pre-processing pipeline. PMG slices were exported at higher native resolution than HC slices, which makes a texture-based shortcut available to the model. Third, pre-split downsampling. HC slices were globally downsampled to match PMG sample size before the data were split, producing artificially balanced evaluation sets, in contrast to the patient-level splits adopted by Zhang et al. (2024). This project replicates the original protocol and then corrects all three errors.

3.3 Experimental setup

To evaluate the methodological validity of Guha et al. (2025) and isolate the contribution of each identified error, experiments were organized into conditions spanning three axes: preprocessing (raw vs. pre-processed), label definition (paper vs. corrected), and downsampling strategy (none, pre-split, post-split). Both ResNet-101 and DenseNet-201 were evaluated under all conditions using 5-fold cross-validation at the patient level. Table 3.3 and Figure 5 provide an overview.

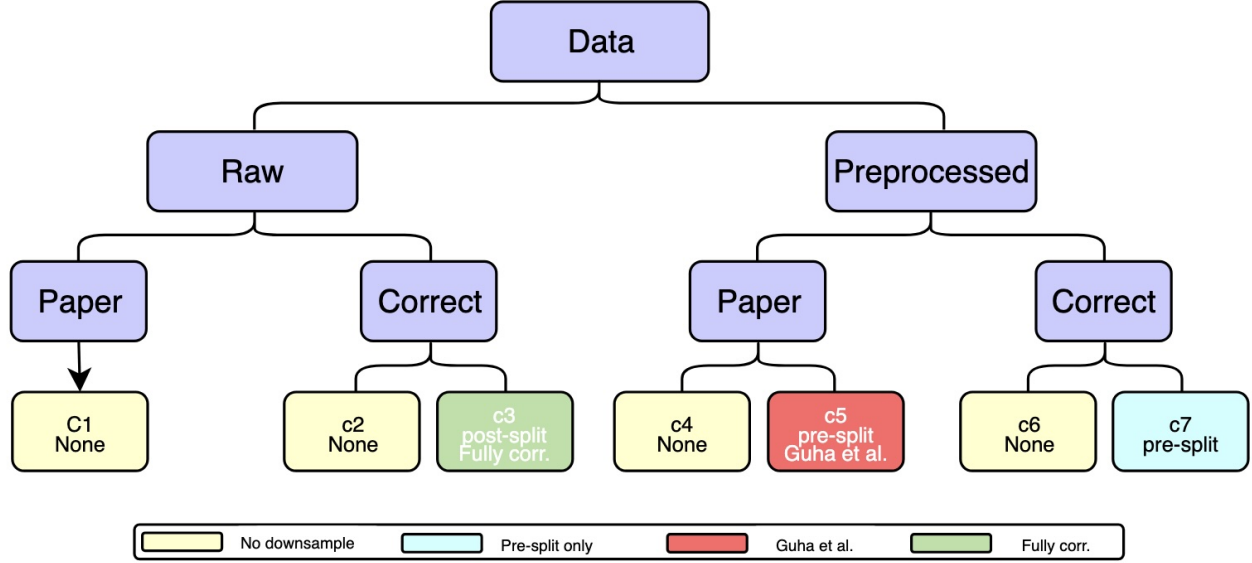


Figure 5: Experimental design tree defining the seven evaluation conditions (C1–C7). The dataset divides into two preprocessing branches, *raw* and *preprocessed*; each branch is split by label definition into a *paper* (following Guha et al., 2025) and a *corrected*. Each leaf C1–C7 is a unique combination of preprocessing, label definition, and downsampling strategy (none, pre-split, or post-split): a condition is therefore fully specified by its path from the root. C5 reproduces Guha et al.’s exact setup (preprocessed, paper labels, pre-split downsampling), whereas C3 is the fully corrected condition (raw, corrected labels, post-split downsampling).

Condition	Preprocessing	Labels	Downsampling
1	Raw	Paper	None
2	Raw	Corrected	None
3	Raw	Corrected	Post-split
4	Preprocessed	Paper	None
5	Preprocessed	Paper	Pre-split
6	Preprocessed	Corrected	None
7	Preprocessed	Corrected	Pre-split

Table 3: Overview of the seven experimental conditions spanning three axes: preprocessing (raw vs. preprocessed), label definition (paper vs. corrected), and downsampling strategy (none, pre-split, post-split).

3.3.1 Data label definition

Following Guha et al. (2025), the *paper’s label definition* treats all 4517 slices attributed to PMG patients as the positive class, regardless of their slice-level annotations, and all 10539 HC slices form the negative class, see patient-level slice distribution in Figure 3a-b. We updated the label definition (*corrected label definition*), restricting the positive class to slices with label = 1 ($n = 2256$). Label = 2 ($n = 1386$) are reclassified as negative examples and pooled with the original HCs, yielding $10539 + 1386 = 11925$ negative slices, see Figure 3c full slice distribution of cases and HC. Label = 3 slices are excluded from corrected conditions. All splits were performed at the patient-level: the 92 unique subjects were partitioned such that all slices from a given patient appear exclusively in one split. A fixed random seed (seed = 42) ensured reproducibility.

3.3.2 Resolution confound and Preprocessing

The resolution confound is a property of the raw data: PMG slices were exported at a substantially higher native resolution than HC slices (see Section 4.1.1), so there is a systematic, class-correlated texture difference encoded in the dataset independent of pathology. Therefore, this confound cannot be removed post hoc and is present in all setups. The preprocessing setup was designed to probe the interaction between resolution confounds and the preprocessing pipeline. Preprocessing was treated as an experimental variable with two options: either raw data or preprocessed data. Holding labels and downsampling fixed while toggling the preprocessing setup on and off isolated the pipeline’s effect on performance.

3.3.3 Downsampling Strategy

The 10539 control slices were globally downsampled to 4517 before being split into training (60%), validation (20%), and test (20%) sets, so that all sets are artificially balanced 1:1 class ratio. Both downsampling strategies use the same operation, randomly discarding HC slices down to the PMG sample size, which is blind to slice information content and not patient-aware. Applied before the split (pre-split), it draws randomly across all HC subjects, so it can discard every slice of some HC patients and it balances the validation and tests as well. Only the training set should be balanced, to stabilise gradient updates

during learning. Validation and test sets should reflect the natural balance so that reported metrics generalise to deployment.

The correct, post-split alternative applies the identical random undersampling operation but only to the training split after the patient-level split, leaving validation and test sets at their natural balance and retains every patient in evaluation.

Then 5-fold cross-validation at the patient level was used to obtain stable performance estimates for condition 3 and 4, to compare with Guha et al. results. Patients were divided by class and split into 5 folds, ensuring approximately equal class representation per fold. Within each fold, 15 % of the non-test patients were held out as a validation set; the models checkpoint with the lowest validation loss was saved and evaluated on the test fold.

3.4 Preprocessing

The suggested preprocessing pipeline described in this section replicates, step-by-step, the procedure reported by Guha et al. Guha et al. (2025) and was applied identically across all experimental condition and both datasets. The pipeline comprises five sequential operations applied to each 2D coronal T1-weighted MRI slice.

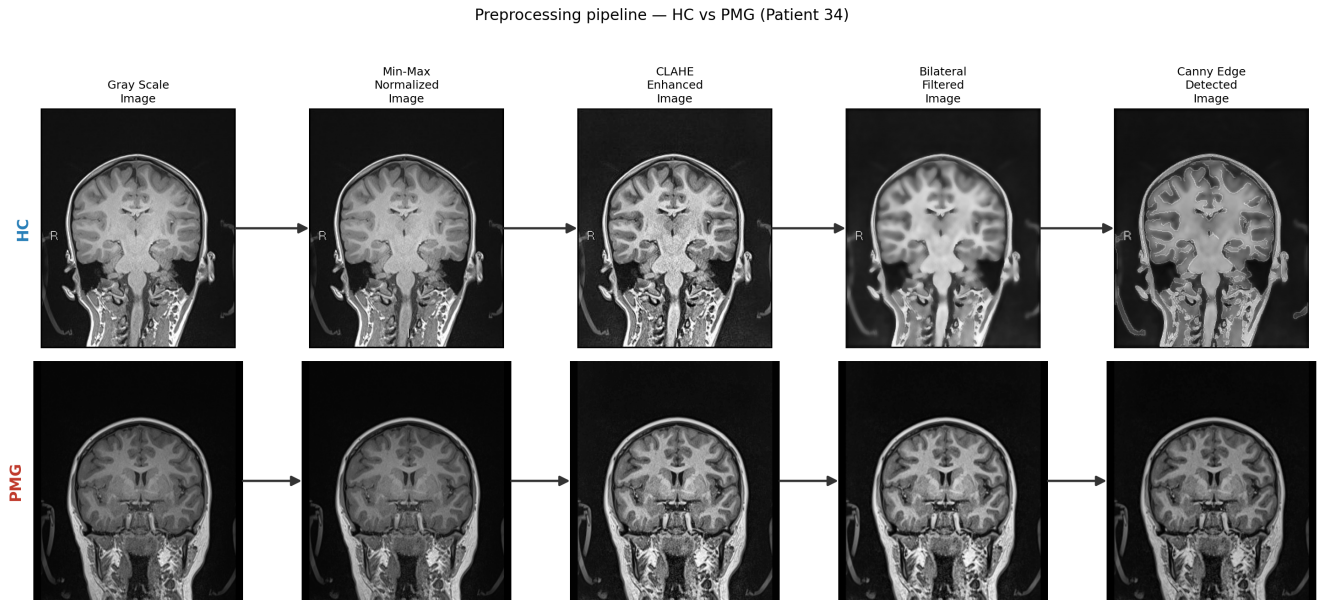


Figure 6: Preprocessing pipeline applied to HC (HC, top) and polymicrogyria (PMG, bottom) slice. From left to right, the five sequential steps replicate Guha et al. (2025): grayscale conversion, min–max intensity normalisation, contrast-limited adaptive histogram equalisation (CLAHE), bilateral filtering, and Canny edge detection. The identical pipeline was applied to both classes and across both datasets.

- (1) **Grayscale conversion.** Each RGB JPEG image was converted to a single-channel grayscale representation. This step reduces computational complexity in the following operations that follow Guha et al. (2025).

- (2) **Min-max normalization.** Pixel intensities within each image were linearly rescaled to the interval $[0, 1]$ on a per-image basis. This standardizes the range across scans acquired under heterogeneous imaging conditions, ensuring that differences in scanner gain or acquisition protocol do not introduce systematic variation into the feature representation presented to the classifier (Guha et al., 2025).
- (3) **Contrast Limited Adaptive Histogram Equalisation (CLAHE).** Local contrast was enhanced using CLAHE (Zuiderveld, 1994), with a clip limit of 2.0 and a grid size of 8×8 pixels. Unlike global histogram equalisation, CLAHE subdivides the image into non-overlapping contextual regions and applies histogram equalisation independently within each grid, enhancing local contrast variation that would otherwise be suppressed by global equalisation methods. The clip limit bounds the amplification applied to any histogram bin, preventing over-amplification of noise characteristic of unconstrained adaptive methods (Zuiderveld, 1994). For MRI of cortical architecture this localized enhancement should improve the visibility of fine structural details such as gyral folding patterns and gray-white matter boundaries (Guha et al., 2025; Rasheed et al., 2023).
- (4) **Bilateral filtering.** Noise reduction was achieved with bilateral filter (Tomasi & Manduchi, 1998), with a kernel size of 9×9 pixels and $\sigma_{color} = \sigma_{space} = 75$. The bilateral filter replaces each pixels intensity with a weighted average of its spatial neighbourhood, where contributions are controlled both by a Gaussian spatial kernel and a Gaussian range kernel (Tomasi & Manduchi, 1998). This dual-weighting mechanism suppresses high-frequency noise in homogeneous regions while preserving sharp transitions at anatomical boundaries – a property important for retaining cortical surface structure (Akar, 2016). Following Guha et al. (2025), who evaluated the bilateral filter against Non-Local Means, Anisotropic Diffusion, and Wavelet Denoising, the bilateral filter was selected as the most effective method for preserving the cortical boundaries and structure.
- (5) **Canny edge detection.** The Canny edge detector computes an edge map (Canny, 1986), with a low threshold 50, a high threshold of 200, and a Sobel aperture size of 3. The canny algorithm applies a multi-stage pipeline comprising Gaussian smoothing, gradient magnitude estimation, non-maximum suppression and hysteresis-based edge tracking, yielding thin, well localized edge contours with reliable noise rejection Canny, 1986. For PMG detection, accurate delineation of subtle cortical irregularities, including irregular gyri and abnormally shallow sulci, depends on precise edge localisation. The Canny detector was selected over the Sobel operation, Scharr operator, and Laplacian of Gaussian filter after evaluation, all of which produced less reliable results on cortical MRI, according to Guha et al. (2025). The resulting binary edge map was blended with the bilaterally filtered image at $\alpha = 0.20$, subtly reinforcing structural boundaries without discarding the underlying intensity information.

3.5 Models and Training

Two pretrained deep convolutional neural network architectures were evaluated, following Guha et al. (2025): ResNet-101 (He et al., 2016) and DenseNet-201 (Huang et al., 2017), both initialized with ImageNet weights. The final fully connected layer of each model was replaced by a two layer classification head: a linear projection to 256 units, ReLU activation, dropout ($p=0.5$), and a final linear layer to a single logit, e.g. PMG vs control, following the head design of Guha et al. (2025). The backbone was frozen and only the classification head was trained, following Guha et al. (2025). Input images were resized to 224×224 pixels and replicated across three channels to match the ImageNet input format. Standard data

augmentation was applied during training: random resized crop, 80-100% of the image area, resized to 224×224 pixels, and random horizontal flip.

Models were trained using the Adam optimizer with a learning rate of 5×10^{-4} , weight decay of 1×10^{-3} , and a batch size of 32, for 20 epochs, with no early stopping. Binary cross-entropy with logits (BCEWithLogitsLoss) was used as the loss function across all conditions. Performance was evaluated using accuracy, precision, recall, F1 score and Cohen’s κ .

3.6 Ablation study

To probe whether trained models rely on spatially localised image features or on global low-level statistics, a black-box occlusion ablation was applied to all retained checkpoints. For each test image, a random square patch with a side length equal to 50% of the shorter image dimension was zeroed out prior to inference, see figure 7. The occluded test set was constructed once per split and then passed through every available checkpoint. Classification metrics were computed on the occluded images and compared to the corresponding unoccluded test performance. A substantial drop in performance under occlusion is consistent with the model attending to specific spatial regions; a negligible drop suggests reliance on global texture or low-level statistics distributed across the image, which would further support the shortcut learning hypothesis.

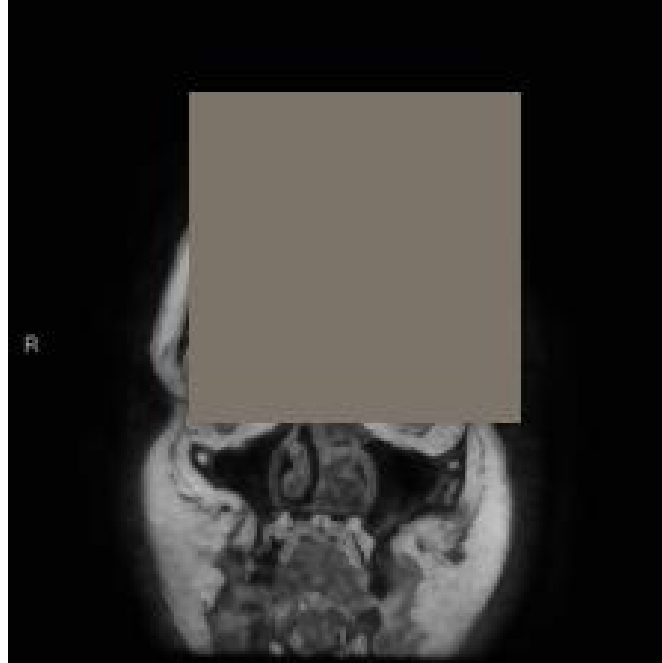


Figure 7: Example occluded image from the black-box ablation study. A square patch with side length equal to 50% of the shorter image dimension (`box_size_frac = 0.5`) is placed at a random location and its pixel intensities set to zero.

4 Results

This section first gives an overview of the dataset, then reports the five-fold cross-validation performance of both models across the seven experimental conditions, the black-box occlusion ablation, and the comparison with the in-house NRU dataset. Three findings stand out (Table 4.3). (1) Preprocessing dom-

inates performance: every preprocessed condition outperformed its raw counterpart (Condition 1 vs. 4 and 2 vs. 6); all preprocessed conditions scored higher than every raw condition for both models. (2) The replication of Guha et al. (2025) (Condition 5) reproduced near-ceiling performance, matching the performance reported by Guha et al. (2025). (3) Correcting the label definition sharply reduced metrics, except accuracy, and the fully corrected condition 3 and condition 2 (raw with corrected, no downsampling) fell to near-chance. Finally, both downsampling strategies increased measured performance, adding a confounding effect rather than cleanly correcting the class imbalance.

4.1 Dataset distribution and characterisation

4.1.1 Label definition

The dataset comprises 15056 slices from 92 patients: 23 PMG cases with three age- and sex-matched HC each, a 3:1 HC-to-PMG ratio at patient level (Zhang et al., 2024). Slices within the PMG cohort are labelled PMG-positive, PMG-negative, or uncertain. Under Guha et al. (2025) the whole PMG cohort forms the positive class (4517, 30%). Under the corrected definition only label = 1 slices form the positive class (2256, 15.9%). The full label distribution is shown in Figure 8 along with Figure 3. The class-balance comparison is in Figure 9; the corrected definition shifts the balance from approximately 30 : 70 to 16 : 84.

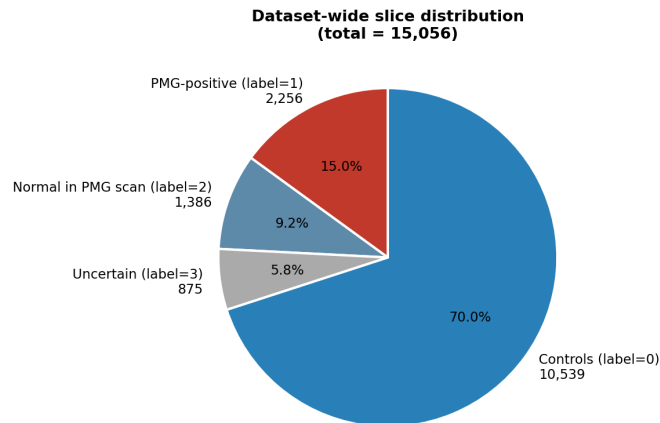


Figure 8: Distribution of all 15056 slices in the PPMR dataset by neuroradiologist annotation. 2256 slices (15.0%) from PMG patients are annotated PMG-positive (label=1); 1386 (9.2%) are normal slices within PMG scans (label=2), 875 (5.8%) are uncertain (label=3), and 10539 (70.0%) are HC slices (label=0). This illustrates that the majority of slices attributed to PMG patients do not contain visible cortical malformation, motivating the corrected label definition.

This difference in label definition results in a 2.0x inflation of the reported positive class, as seen in Figure 9, and introduces noise into the training signal.

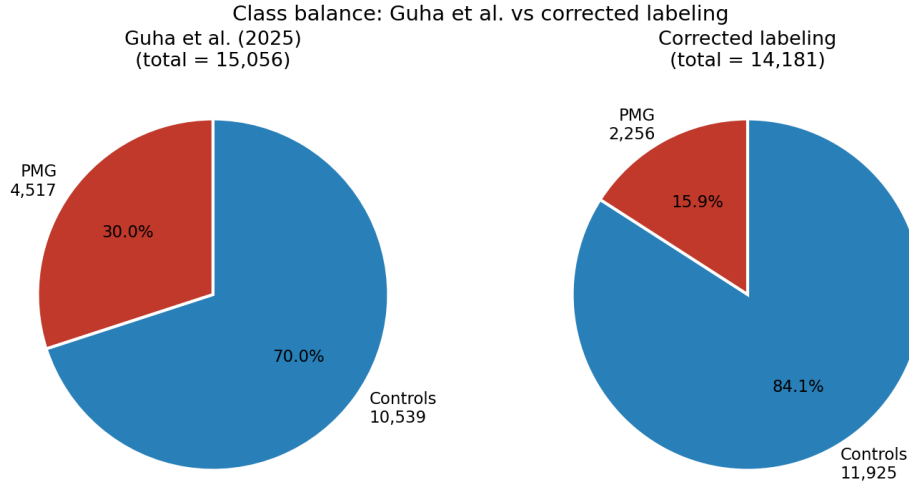


Figure 9: Class balance under the two labeling schemes. **Left:** Guha et al. (2025), where all 4517 slices from PMG patients form the positive class (30.0%) against 10539 controls (70.0%). **Right:** the corrected scheme, which restricts the positive class to the 2256 PMG-positive (label=1) slices (15.9%) and pools label=2 slices with the controls to 11926 (84.1%). Correcting the labels roughly halves the apparent positive-class prevalence, revealing the true class imbalance.

4.1.2 Resolution confound

PMG slices were acquired at a much higher native resolution, median 1508×1727 pixels, than HC slices, 260×260 to 512×512 pixels, a roughly six-fold difference (Figure 11). This allows for differing effect of the preprocessing pipeline on high- compared to low-resolution slices, which is qualitatively shown in 10. Resizing all images to the 224×224 model input therefore downscales PMG slices by $6 - 7\times$, but HC slices by only $1.1 - 2.2\times$, further enhancing the class-correlated difference in spatial-frequency and texture.

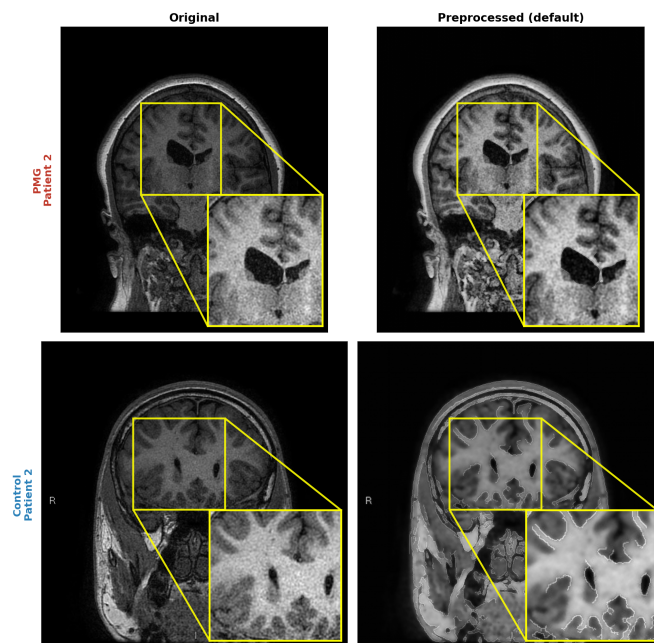


Figure 10: Single-subject detail of the preprocessing pipeline, original (left) versus preprocessed (right), with a magnified inset of the cortical region. **Top:** a PMG patient (Patient 2). **Bottom:** a HC (Patient 2).

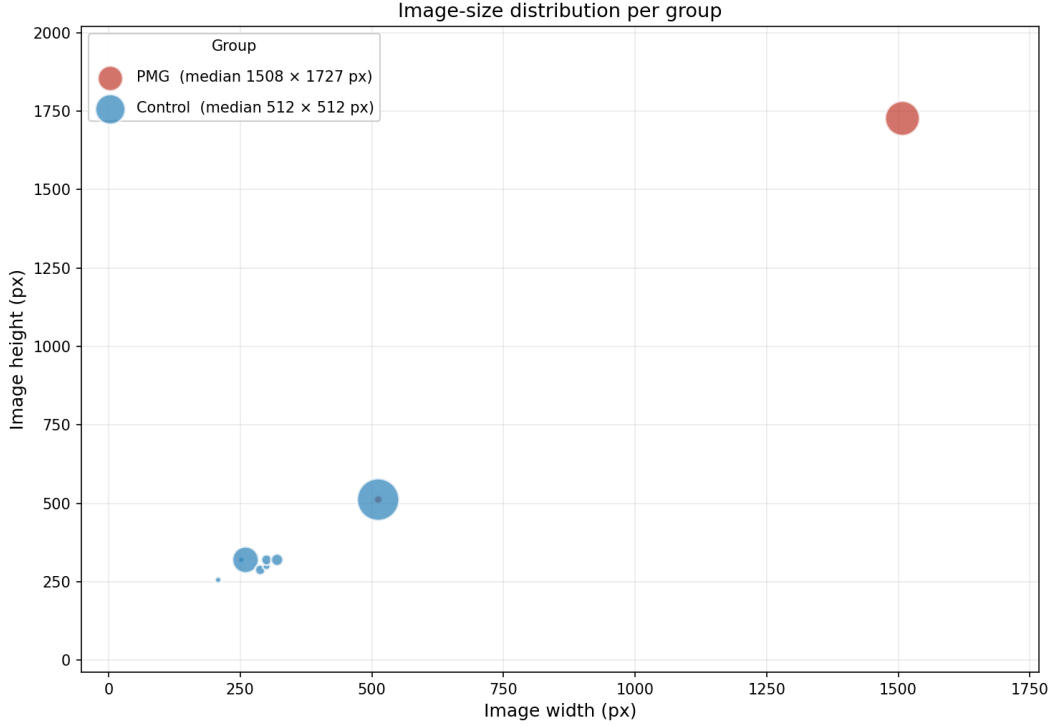


Figure 11: Image-size distribution by group. Each marker is a distinct width $\times$ height combination, with the circles area proportional to the number of slices at that resolution. PMG-patient images cluster at a much higher resolution (median 1508×1727 px) than healthy-control images median 512×512 px). This systematic, class-correlated resolution gap produces a low-level texture confound that a classifier can exploit as a shortcut, independent of any cortical pathology.

The resolution confound is present in all seven experimental conditions, as it is a property of the dataset. Correcting for it requires the original DICOM files to resample volumes into common voxel space.

4.2 Classification Performance Across Experimental Conditions

Table 3.3 defines the seven experimental conditions and Table 4.3 reports five-fold cross-validation performance for both Resnet-101 and DenseNet-201 across all conditions.

Table 4: Five-fold cross-validation results (mean $\pm$ std) for ResNet101 and DenseNet201 across the seven experimental conditions defined in Table 3.3. Metrics: Accuracy (Acc), Precision (Prec), Recall (Rec), F1-score (F1), and Cohen’s κ . **Orange** = Guha et al. replication (Condition 5). **Green** = fully corrected protocol (Condition 3).

C	Preprocessing	Labels	Downsampling	ResNet101					DenseNet201				
				Acc	Prec	Rec	F1	κ	Acc	Prec	Rec	F1	κ
1	Raw	Paper	None	.850 $\pm$.042	.849 $\pm$.054	.621 $\pm$.131	.708 $\pm$.091	.613 $\pm$.107	.843 $\pm$.056	.874 $\pm$.044	.567 $\pm$.209	.665 $\pm$.174	.577 $\pm$.178
2	Raw	Corrected	None	.839 $\pm$.020	.515 $\pm$.100	.206 $\pm$.097	.275 $\pm$.100	.210 $\pm$.082	.850 $\pm$.033	.590 $\pm$.204	.260 $\pm$.129	.343 $\pm$.141	.276 $\pm$.129
3	Raw	Corrected	Post-split	.806 $\pm$.041	.424 $\pm$.049	.520 $\pm$.170	.449 $\pm$.071	.336 $\pm$.079	.832 $\pm$.034	.450 $\pm$.169	.408 $\pm$.191	.414 $\pm$.172	.322 $\pm$.169
4	Preprocessed	Paper	None	.980 $\pm$.032	.991 $\pm$.006	.945 $\pm$.103	.965 $\pm$.058	.951 $\pm$.079	.982 $\pm$.034	.995 $\pm$.005	.947 $\pm$.110	.967 $\pm$.061	.955 $\pm$.084
5	Preprocessed	Paper	Pre-split	.971 $\pm$.055	.995 $\pm$.004	.950 $\pm$.105	.969 $\pm$.059	.943 $\pm$.108	.970 $\pm$.058	.995 $\pm$.005	.948 $\pm$.113	.967 $\pm$.064	.940 $\pm$.115
6	Preprocessed	Corrected	None	.917 $\pm$.034	.755 $\pm$.096	.711 $\pm$.143	.728 $\pm$.105	.679 $\pm$.124	.919 $\pm$.033	.765 $\pm$.080	.720 $\pm$.118	.739 $\pm$.091	.691 $\pm$.110
7	Preprocessed	Corrected	Pre-split	.881 $\pm$.078	.915 $\pm$.038	.835 $\pm$.149	.868 $\pm$.097	.760 $\pm$.155	.899 $\pm$.085	.922 $\pm$.020	.867 $\pm$.166	.887 $\pm$.106	.797 $\pm$.168

* All values are mean $\pm$ standard deviation across 5 folds. Seed fixed at 42 throughout.

4.2.1 Effect of preprocessing

Applying the preprocessing pipeline to paper labeled data (Condition 1 vs. 4) substantially increases performance for both models across all metrics (Table 4). As a result of the preprocessing pipeline applying fixed parameters regardless of a slice’s native resolution, it does not act uniformly across the two cohorts.

4.2.2 Effect of label definitions

Correcting the label definition on raw and preprocessed data (Condition 1 vs. 2 and Condition 4 vs. 6, respectively) reduces precision and recall markedly for both models, while accuracy declines more modestly (Table 4).

4.2.3 Effect of pre-split vs. post-split downsampling

Because the post-split downsampling only appears in the raw conditions and pre-split only in preprocessed ones, no single pair isolates pre- against post-split directly; each is read against its own no-downsampling baseline (Table 4). Post-split downsampling (Condition 2 vs. 3) raises recall, F1 and Cohen’s κ , while precision drops and accuracy is largely unchanged; pre-split downsampling (Condition 6 vs. 7) raises precision, recall, F1 and Cohen’s κ together with a slight fall in accuracy. These effects are interpreted in Section 5.3.

4.3 Occlusion Abaltion study

Table 5: Black-box occlusion ablation, mean $\pm$ std over the five best-epoch checkpoints. Metrics are positive-class (PMG) values computed per checkpoint. **Orange** = Guha et al. replication (C5; preprocessed + paper labels + pre-split). **Green** = fully corrected protocol (C3; raw + corrected labels + post-split).

Condition	Model	Acc	Prec	Rec	Spec	F1	κ
C5	ResNet-101	.878 $\pm$.010	.520 $\pm$.036	.591 $\pm$.165	.921 $\pm$.018	.546 $\pm$.098	.477 $\pm$.101
	DenseNet-201	.801 $\pm$.096	.353 $\pm$.092	.401 $\pm$.302	.860 $\pm$.155	.319 $\pm$.064	.222 $\pm$.039
C3	ResNet-101	.870 $\pm$.005	.519 $\pm$.145	.104 $\pm$.067	.984 $\pm$.012	.163 $\pm$.090	.127 $\pm$.074
	DenseNet-201	.687 $\pm$.274	.181 $\pm$.184	.171 $\pm$.278	.763 $\pm$.355	.073 $\pm$.076	-.036 $\pm$.050

* Occlusion box side = 50% of $\min(h, w)$ (25% of the image area). Values are mean $\pm$ std across the five best-epoch checkpoints.

† The high variance for DenseNet-201/C3 reflects one fold collapsing to near-chance performance, indicating instability under the corrected protocol.

4.4 Comparison between in-house dataset and PPMR dataset

The models were trained and evaluated on the independent NRU clinical dataset using the same five-fold patient-level cross-validation. Performance is far below the PPMR figures: both models reach only ~ 0.60 – 0.62 accuracy and near-zero agreement (Cohen’s $\kappa \approx 0.09$), close to chance (Table 6). Per-fold results are given in Appendix 18.

Table 6: NRU in-house dataset: five-fold cross-validation results (mean $\pm$ std). Per-fold values are given in Appendix 18.

Model	Acc	Prec	Rec	F1	κ
ResNet-101	.618 $\pm$.046	.655 $\pm$.023	.831 $\pm$.109	.730 $\pm$.056	.089 $\pm$.045
DenseNet-201	.599 $\pm$.061	.656 $\pm$.030	.761 $\pm$.124	.702 $\pm$.066	.087 $\pm$.090

5 Discussion

5.1 Label definition

Comparing the label definition of Guha et al. (2025) against the corrected definition, the paper definition clearly produces a stronger performance. However, this is exactly the underlying problem we are trying to expose. Under the paper definition, every slice from a PMG patient is treated as PMG-positive, even though many such slices do not contain visible cortical malformation. The classifier will be rewarded for separating the two patient cohorts on any available signal, pathology or not.

The results show this as well. On the preprocessed data, moving from paper definition to the corrected one (Condition 4 vs. 6, see Table 4) all metrics drop for both models (except accuracy). This same pattern appears on the raw data (Condition 1 vs. 2, see Table 4). When the easy cohort-level signal is removed by moving to corrected label definition, the models lose most of their discriminative power. This is a signature sign of shortcut learning: high scores under the paper definition reflect, in large part, the separation of two cohorts on non-pathology signal rather than the detection of real pathology.

Accuracy, however, is the only metric that does not decline significantly across comparisons (see for example condition 4 vs. 6). This is expected: with a majority of HC slices, a model that defaults towards the HC class retains high accuracy even as precision, recall and other metrics for the minority PMG class drop. Accuracy alone therefore masks the collapse that precision, recall and others reveal.

5.2 Resolution confound and preprocessing effects

The main driver of the shortcut learning is a systematic difference in image resolution between the two cohorts. As characterised in Section 4.1.2, PMG slices have a median resolution of 1508×1727 pixels. Whereas HC range from 260×260 to 512×512 pixels, with a median of 512×512 pixels — a roughly six-fold difference in resolution. Resizing every slice to 224×224 pixels downscales the PMG slices by about six- to seven-fold but the HC slices by only $1.1 - 2.2\times$ (Figure 11). Because the two classes are resized by very different factors, the resizing step itself imprints a class-correlated difference on the data. This resolution difference is a property of the raw data and cannot be corrected post hoc once the slices have been exported from the PACS system as JPEGs. The fixed-size resizing is a property of using the model input of size 224×224 . We contacted Zhang et al. (2024) for the source DICOMs in order to resample all scans to a common voxel spacing, but they stopped responding when confronted with data inconsistencies.

The preprocessing pipeline amplifies rather than harmonises the resolution confound. CLAHE and the bilateral filter operate with fixed kernel and tile sizes regardless of a slice’s native resolution, an identical kernel spans a much smaller fraction of anatomy on a high-resolution PMG slice than on a lower-

resolution HC slice. The enhancement is therefore subtle on PMG slice but has a significantly stronger effect on the HC slices (Figure 10), so the pipeline acts unequally on the two classes and deepens, rather than removes, the resolution-correlated difference.

The evidence for this is given in three parts. (1) Preprocessing produces a large and fairly uniform jump in performance under the paper and corrected labels (Condition 1 vs. 4, and Condition 2 vs. 6, Table 4). A jump of this magnitude is difficult to attribute to better delineation of subtle cortical features. It is more likely that the preprocessing increased performance by enhancing the existing differences between the cohorts. (2) The single-subject detail (Figure 10) shows CLAHE and bilateral filtering acting more strongly on the lower-resolution HC than on the PMG, illustrating the unequal treatment that the aggregate metrics imply. (3) The occlusion ablation (Table 5), shows that masking 25% of the image area produces only a modest drop in metric performance, indicating reliance on globally distributed texture and spatial frequency. This point will be further discussed in Section 5.4.

5.3 Data splitting

Guha et al. (2025) acknowledge the 3:1 natural class imbalance and address it by globally downsampling the 10539 control slices to 4517 before splitting the data into training, validation, and test sets. This procedure may introduce data leakage. The 6022 discarded HC slices may have belonged to patients who would otherwise have been represented in the validation or test set; after downsampling, those patients are absent or underrepresented in evaluation, so the composition of the validation and test sets is determined by the downsampling step together with the split step rather than by the split alone. Because balancing is applied globally before the split, the validation and test sets are themselves balanced at 1:1 ratio, which does not represent the true clinical prevalence of PMG. This inflates the evaluation metrics relative to a real-world deployment in which PMG is rare (Buda et al., 2018; van den Goorbergh et al., 2022). The downsampling is also content-blind, it is a random draw, so it may discard information-dense slices and retain slices with no information. A position-aware or content-aware sampler would be preferable.

The corrective post-split downsampling strategy fixes the leakage but not the content-blindness. It applies the same random, content-blind undersampling of HC slices, differing only in that it does so within the training set. Therefore, it never removes a patient from evaluation and never balances the validation or test sets, but it can still retain uninformative slices and discard informative ones in training. It also imposes an implicit 1:1 training prior that mismatches the imbalanced validation and test sets. For this reason post-split downsampling is not a clean correction of Guha et al. (2025) pre-split downsampling. The results show that moving from no downsampling to post-split downsampling (Condition 3 vs. 2, see Table 4) increases performance slightly. This gain in metric performance should be seen as a training-stability effect rather than as evidence that the post-split strategy recovers genuine pathology signal.

5.4 Ablation study

To evaluate whether the models depend on pathology/morphology of PMG or on globally distributed image properties, we occluded a square patch covering 25% of the image area (box side equal to 50% of $\min(h, w)$) and re-evaluated every checkpoint for condition 3 and 5 (Table 5).

For the Guha replication (Condition 5), occlusion left accuracy and specificity close to their unoccluded values but sharply reduced positive-class recall and precision (Table 5). The retained accuracy and high

specificity show that the models still identify the majority HC class after a quarter of the image is removed, which is consistent with reliance on globally distributed low-level statistics rather than on localised pathology and an imbalanced dataset. The fall in recall, however, shows that the masked region also carried part of the positive-class, this is suggestive of, but not conclusive proof of shortcut learning.

Under the fully corrected protocol (Condition 3) models retain little robust positive-class signal. ResNet-101 keeps high accuracy and specificity, see Table 5. But the recall collapses to 0.104: under occlusion it abandons positive predictions almost entirely and defaults to the majority class.

These findings gives two cautions about the models behaviours. (1) An occlusion ablation is suggestive rather than definitive. The masked patch is large and applied per slice so some recall drop is expected for any model and does not by itself prove reliance on global level textures, shortcut learning. (2) The metrics are dominated by the strong class imbalance — accuracy and specificity are kept high by the majority HC class — so recall is the most informative column here. Taken together with the cross-validation and resolution analyses, the ablation is consistent with, though not stand alone proof of, reliance on globally distributed low-level texture rather than detection of genuine cortical pathology.

5.5 NRU dataset

The NRI dataset provides a useful counterfactual to the PPMR dataset. It follows the same protocol as Guha et al. (2025), every slice from a PMG patient is labeled as PMG-positive and every slice from a HC patient is PMG-negative. Unlike PPMR, it carries no class-correlated resolution confound, since all scans were acquired as volumes at a common voxel space and sliced uniformly, so PMG and HC slices share the same resolution. Under this protocol both models had mediocre performance, with Cohen’s κ falling to near-chance (approx 0.09, see Table 6), results far below the conditions 1, 4, and 5, which are the paper label definition conditions. This is the expected performance without the resolution shortcut available to the model. The NRU result therefore corroborates the central findings — that the near-perfect PPMR number reflect the resolution confound rather than genuine PMG detection — by showing that the same protocol, stripped only of resolution confound, performs little better than chance.

5.6 Future work

Two directions should be further investigated. First, the resolution confound should be characterised in the intensity domain to complement the spatial-texture evidence of Section 4.2.1 and 4.1.2. We propose comparing per-class intensity histograms at three stages — the raw JPEG slice, the output of the preprocessing pipeline, and the slice after resizing to 224×224 pixels — to test directly whether preprocessing and resizing harmonise the two classes or, as the qualitative evidence suggests, separate them further. If the histograms of the PMG and HC classes diverge more after preprocessing and resizing than before, that would confirm in the intensity domain what the texture and ablation analyses imply.

Second, the slice-level framing should be replaced with a volumetric one. Using the volumetric NRU data, a 3D CNN or a transformer could be trained on whole volumes rather than on individual 2D slices. A volumetric model removes the PMG-negative slice problem at source — there is no per-slice label to mislabel — and would avoid the same per-slice resolution artefact that drives present shortcut learning, this could offer a more honest test of whether deep learning can detect PMG from genuine cortical morphology and pathology.

6 Conclusion

This project set out to reproduce the near-perfect performance reported by Guha et al. (2025), and this was achieved. Correcting for three identified methodological errors then reduced performance metrics substantially – with one exception. Moving from no downsampling to post-split downsampling under, corrected labels (Condition 2 vs. 3) increased metrics significantly, see Table 4. The post-split balancing method is difficult to classify as either methodological upgrade or a milder form of the same flawed balancing as the pre-split method. As it relies on the same random, content-blind undersampling of HC slices.

The occlusion ablation is consistent with reliance on global image texture, e.g. shortcut learning, rather than genuine detection of PMG pathology signals. Therefore, as a stand alone analysis it cannot prove nor disprove the shortcut learning theory.

Two methodological lessons can be derived from the results and discussion: (1) Trustworthy, reproducible results requires label fidelity, control of class-correlated confounds, and an honest patient-level split. Where any class balancing applied only to training set, after the split, and never to the validation or test sets. Ignoring these lessons is enough to produce performance that looks excellent but means nothing. Second, which underlies the first, is that a lack of domain knowledge was clearly at play. The errors corrected here arise less from modeling choices than from just not knowing the data – how each slice was labeled, the resolution at which the cohort were exported, and how the classes were balanced before evaluation. Knowing the data, understanding how it was produced, and following sound data practice are prerequisites for results that can be reproduced and trusted; without that, a model can report near-perfect scores while learning nothing about the pathology it is meant to detect.

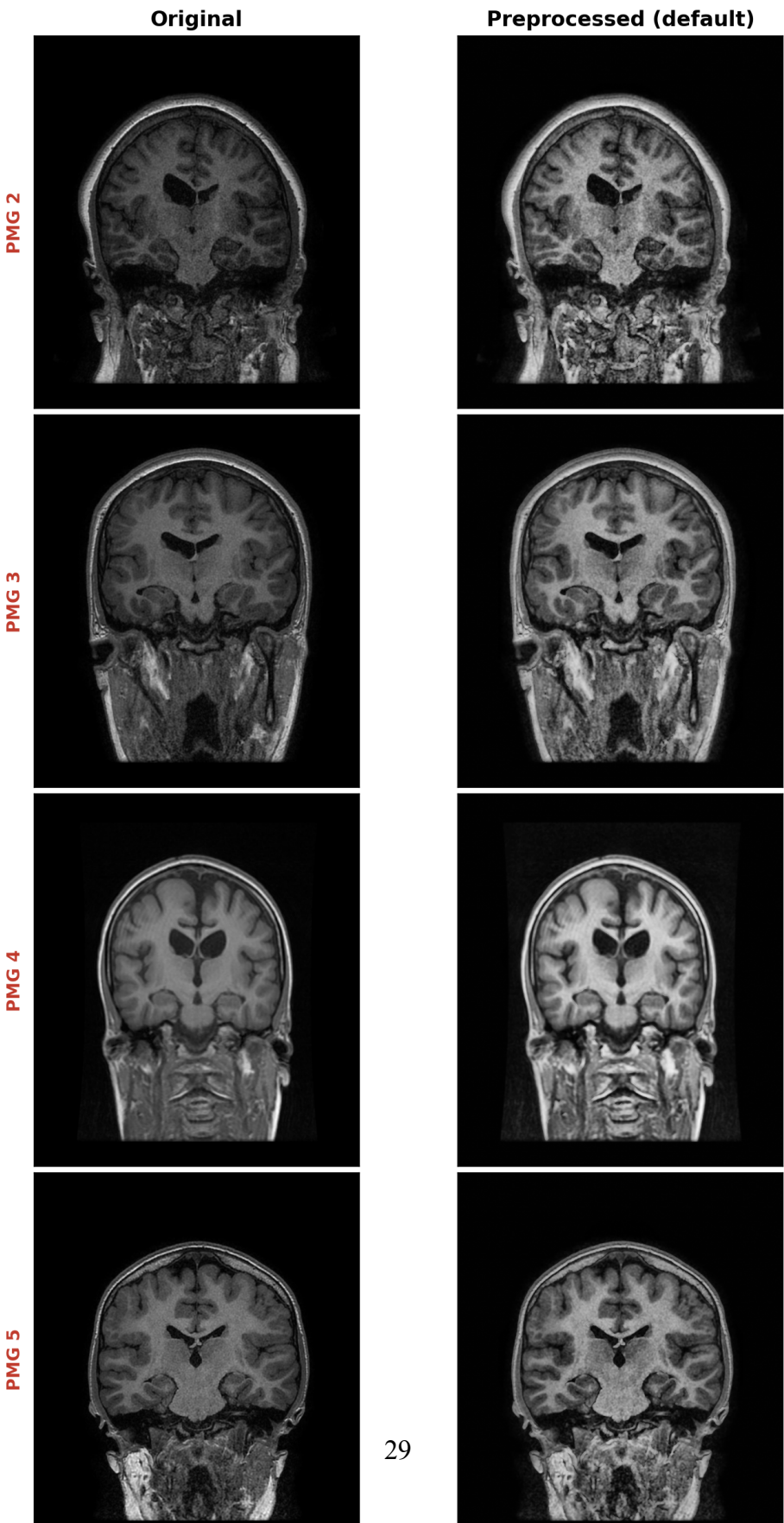
This is reinforced by the independent NRU dataset. Evaluated under the same per-slice protocol as Guha et al. (2025), but free of the class-correlated resolution confound, both models performed little better than chance. Exactly what would have been expected of a small dataset like PPMR without the resolution confound.

Code and Data Availability

The full preprocessing, training, cross-validation, and ablation code, together with the configuration files needed to reproduce all experiments in this report, is available at <https://github.com/HolgeMax/PMG>. The PPMR dataset is publicly available (Zhang et al., 2024); the NRU clinical dataset cannot be shared owing to patient-confidentiality restrictions.

A **Supplementary figures**

PMG group — Original vs Preprocessed (default)



Control group — Original vs Preprocessed (default)

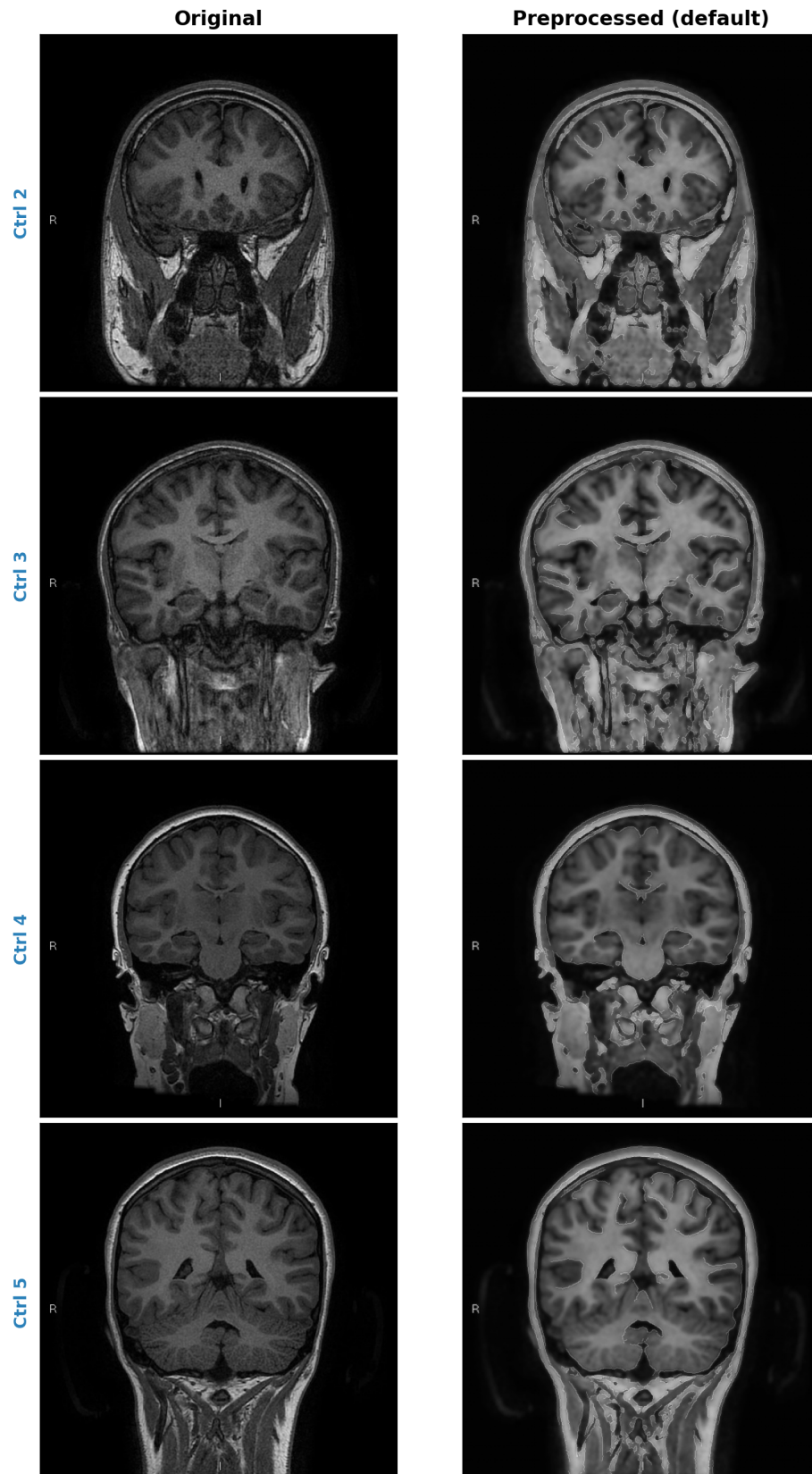


Figure 13: Preprocessing pipeline applied to a representative healthy-control slice.

B Ablation study — per-checkpoint results

Table 7: Per-checkpoint occlusion ablation — C5, ResNet-101.

Checkpoint	Acc	Prec	Rec	Spec	F1	κ
resnet101_preprocessed_fold1_best	.865	.467	.321	.946	.381	.308
resnet101_preprocessed_fold1_final	.840	.441	.891	.833	.590	.504
resnet101_preprocessed_fold2_best	.875	.510	.724	.897	.599	.527
resnet101_preprocessed_fold2_final	.797	.382	.926	.778	.541	.438
resnet101_preprocessed_fold3_best	.893	.566	.714	.919	.631	.570
resnet101_preprocessed_fold3_final	.887	.555	.621	.926	.586	.520
resnet101_preprocessed_fold4_best	.879	.524	.639	.914	.576	.506
resnet101_preprocessed_fold4_final	.858	.470	.785	.869	.588	.509
resnet101_preprocessed_fold5_best	.879	.530	.557	.927	.543	.474
resnet101_preprocessed_fold5_final	.865	.451	.207	.963	.284	.220

Table 8: Per-checkpoint occlusion ablation — C5, DenseNet-201.

Checkpoint	Acc	Prec	Rec	Spec	F1	κ
densenet201_preprocessed_fold1_best	.855	.359	.162	.957	.223	.155
densenet201_preprocessed_fold1_final	.855	.359	.162	.957	.223	.155
densenet201_preprocessed_fold2_best	.830	.330	.305	.908	.317	.220
densenet201_preprocessed_fold2_final	.853	.385	.236	.944	.293	.216
densenet201_preprocessed_fold3_best	.633	.249	.915	.592	.392	.237
densenet201_preprocessed_fold3_final	.863	.453	.308	.945	.367	.293
densenet201_preprocessed_fold4_best	.815	.326	.406	.875	.361	.255
densenet201_preprocessed_fold4_final	.815	.326	.406	.875	.361	.255
densenet201_preprocessed_fold5_best	.871	.500	.218	.968	.303	.244
densenet201_preprocessed_fold5_final	.745	.310	.798	.737	.447	.320

Table 9: Per-checkpoint occlusion ablation — C3, ResNet-101.

Checkpoint	Acc	Prec	Rec	Spec	F1	κ
resnet101_raw_fold1_best	.862	.304	.056	.981	.094	.057
resnet101_raw_fold1_final	.835	.231	.119	.941	.157	.076
resnet101_raw_fold2_best	.872	.508	.178	.974	.263	.210
resnet101_raw_fold2_final	.870	.488	.215	.967	.298	.238
resnet101_raw_fold3_best	.869	.479	.178	.971	.259	.204
resnet101_raw_fold3_final	.873	.530	.141	.982	.222	.178
resnet101_raw_fold4_best	.874	.667	.048	.996	.089	.073
resnet101_raw_fold4_final	.862	.397	.127	.971	.193	.139
resnet101_raw_fold5_best	.874	.639	.061	.995	.111	.091
resnet101_raw_fold5_final	.862	.424	.191	.962	.263	.199

Table 10: Per-checkpoint occlusion ablation — C3, DenseNet-201.

Checkpoint	Acc	Prec	Rec	Spec	F1	κ
densenet201_raw_fold1_best	.751	.032	.032	.858	.032	-.111
densenet201_raw_fold1_final	.564	.073	.204	.617	.108	-.101
densenet201_raw_fold2_best	.871	.500	.005	.999	.010	.008
densenet201_raw_fold2_final	.871	.500	.003	1.000	.005	.004
densenet201_raw_fold3_best	.211	.102	.655	.145	.176	-.060
densenet201_raw_fold3_final	.152	.110	.782	.058	.192	-.044
densenet201_raw_fold4_best	.867	.158	.008	.994	.015	.003
densenet201_raw_fold4_final	.768	.087	.085	.869	.086	-.047
densenet201_raw_fold5_best	.733	.113	.156	.819	.132	-.021
densenet201_raw_fold5_final	.696	.108	.186	.772	.136	-.032

C Cross-validation — per-fold results

Table 11: Per-fold five-fold cross-validation — C1 (Raw, Paper, None).

Fold	Acc	Bal. Acc	Prec	Rec	Spec	F1	κ
<i>ResNet-101</i>							
1	.880	.840	.829	.744	.936	.784	.701
2	.893	.822	.859	.681	.963	.760	.692
3	.865	.819	.862	.690	.948	.767	.674
4	.807	.748	.774	.576	.919	.661	.530
5	.804	.699	.921	.414	.984	.571	.466
Mean	.850	.785	.849	.621	.950	.708	.613
SD	.042	.060	.054	.131	.025	.091	.107
<i>DenseNet-201</i>							
1	.889	.852	.843	.763	.941	.801	.724
2	.868	.745	.941	.501	.990	.654	.582
3	.853	.808	.828	.683	.933	.749	.646
4	.861	.808	.887	.656	.959	.754	.660
5	.747	.609	.872	.234	.984	.369	.271
Mean	.843	.764	.874	.567	.961	.665	.577
SD	.056	.095	.044	.209	.025	.174	.178

Table 12: Per-fold five-fold cross-validation — C2 (Raw, Corrected, None).

Fold	Acc	Bal. Acc	Prec	Rec	Spec	F1	κ
<i>ResNet-101</i>							
1	.850	.610	.451	.276	.945	.342	.263
2	.867	.616	.389	.291	.941	.333	.261
3	.818	.531	.638	.072	.990	.130	.095
4	.831	.609	.583	.259	.959	.359	.278
5	.828	.554	.512	.134	.973	.213	.151
Mean	.839	.584	.515	.206	.962	.275	.210
SD	.020	.039	.100	.097	.020	.100	.082
<i>DenseNet-201</i>							
1	.849	.634	.454	.335	.933	.385	.301
2	.902	.582	.859	.168	.996	.281	.253
3	.834	.653	.599	.361	.944	.451	.360
4	.854	.661	.698	.356	.965	.471	.397
5	.814	.523	.339	.079	.968	.129	.067
Mean	.850	.611	.590	.260	.961	.343	.276
SD	.033	.057	.204	.129	.024	.141	.129

Table 13: Per-fold five-fold cross-validation — C3 (Raw, Corrected, Post-split).

Fold	Acc	Bal. Acc	Prec	Rec	Spec	F1	κ
<i>ResNet-101</i>							
1	.814	.741	.401	.640	.842	.493	.387
2	.850	.782	.408	.694	.871	.514	.433
3	.740	.654	.365	.518	.791	.428	.266
4	.800	.680	.457	.490	.870	.473	.350
5	.825	.600	.490	.256	.944	.336	.247
Mean	.806	.692	.424	.520	.864	.449	.336
SD	.041	.072	.049	.170	.055	.071	.079
<i>DenseNet-201</i>							
1	.832	.679	.416	.465	.892	.439	.341
2	.866	.694	.421	.471	.917	.444	.368
3	.800	.715	.475	.578	.852	.522	.397
4	.865	.703	.703	.449	.958	.548	.473
5	.795	.512	.234	.079	.945	.118	.033
Mean	.832	.661	.450	.408	.913	.414	.322
SD	.034	.084	.169	.191	.043	.172	.169

Table 14: Per-fold five-fold cross-validation — C4 (Preprocessed, Paper, None).

Fold	Acc	Bal. Acc	Prec	Rec	Spec	F1	κ
<i>ResNet-101</i>							
1	.997	.994	1.000	.989	1.000	.994	.992
2	.997	.998	.991	.999	.997	.995	.993
3	.996	.997	.986	1.000	.993	.993	.990
4	.987	.985	.984	.977	.993	.981	.971
5	.923	.880	.994	.761	.998	.862	.810
Mean	.980	.971	.991	.945	.996	.965	.951
SD	.032	.051	.006	.103	.003	.058	.079
<i>DenseNet-201</i>							
1	.999	.999	1.000	.998	1.000	.999	.999
2	.998	.999	.992	1.000	.997	.996	.995
3	.996	.997	.988	1.000	.994	.994	.991
4	.993	.991	.992	.986	.996	.989	.984
5	.921	.876	1.000	.751	1.000	.858	.805
Mean	.982	.972	.995	.947	.998	.967	.955
SD	.034	.054	.005	.110	.002	.061	.084

Table 15: Per-fold five-fold cross-validation — C5 (Preprocessed, Paper, Pre-split).

Fold	Acc	Bal. Acc	Prec	Rec	Spec	F1	κ
<i>ResNet-101</i>							
1	.997	.997	1.000	.994	1.000	.997	.994
2	.995	.996	.991	.999	.993	.995	.991
3	.998	.998	.997	1.000	.996	.998	.997
4	.992	.992	.990	.994	.989	.992	.984
5	.874	.879	.997	.761	.998	.863	.750
Mean	.971	.972	.995	.950	.995	.969	.943
SD	.055	.052	.004	.105	.004	.059	.108
<i>DenseNet-201</i>							
1	.998	.998	1.000	.997	1.000	.998	.997
2	.996	.997	.991	1.000	.993	.995	.992
3	.996	.996	.993	1.000	.992	.996	.992
4	.993	.992	.989	.997	.988	.993	.985
5	.866	.872	1.000	.745	1.000	.854	.735
Mean	.970	.971	.995	.948	.995	.967	.940
SD	.058	.055	.005	.113	.005	.064	.115

Table 16: Per-fold five-fold cross-validation — C6 (Preprocessed, Corrected, None).

Fold	Acc	Bal. Acc	Prec	Rec	Spec	F1	κ
<i>ResNet-101</i>							
1	.921	.834	.725	.713	.955	.719	.673
2	.946	.914	.716	.872	.956	.786	.756
3	.946	.897	.886	.818	.976	.851	.818
4	.907	.804	.812	.642	.967	.717	.663
5	.865	.725	.636	.512	.939	.568	.488
Mean	.917	.835	.755	.711	.958	.728	.679
SD	.034	.076	.096	.143	.014	.105	.124
<i>DenseNet-201</i>							
1	.925	.853	.726	.754	.953	.740	.696
2	.959	.909	.807	.844	.974	.825	.802
3	.933	.885	.833	.809	.962	.821	.780
4	.905	.794	.816	.619	.969	.704	.649
5	.871	.753	.642	.573	.933	.606	.529
Mean	.919	.839	.765	.720	.958	.739	.691
SD	.033	.065	.080	.118	.016	.091	.110

Table 17: Per-fold five-fold cross-validation — C7 (Preprocessed, Corrected, Pre-split).

Fold	Acc	Bal. Acc	Prec	Rec	Spec	F1	κ
<i>ResNet-101</i>							
1	.898	.895	.943	.834	.956	.885	.794
2	.921	.924	.877	.939	.909	.907	.839
3	.950	.949	.950	.961	.936	.955	.898
4	.885	.889	.933	.854	.924	.892	.771
5	.748	.749	.870	.587	.911	.701	.497
Mean	.881	.881	.915	.835	.927	.868	.760
SD	.078	.078	.038	.149	.019	.097	.155
<i>DenseNet-201</i>							
1	.938	.939	.911	.961	.917	.936	.876
2	.933	.928	.928	.905	.952	.916	.860
3	.967	.963	.946	.998	.929	.971	.933
4	.904	.905	.933	.889	.922	.910	.806
5	.753	.754	.893	.579	.930	.703	.507
Mean	.899	.898	.922	.867	.930	.887	.797
SD	.085	.083	.020	.166	.013	.106	.168

D Cross-validation — NRU in-house dataset

Table 18: Per-fold five-fold cross-validation results on the NRU in-house dataset.

Fold	Acc	Prec	Rec	F1	κ
<i>ResNet-101</i>					
1	.542	.625	.643	.634	.022
2	.628	.670	.838	.745	.091
3	.639	.661	.868	.750	.143
4	.662	.680	.916	.780	.114
5	.618	.637	.890	.743	.076
Mean	.618	.655	.831	.730	.089
SD	.046	.023	.109	.056	.045
<i>DenseNet-201</i>					
1	.532	.609	.674	.640	-.025
2	.583	.663	.722	.691	.051
3	.679	.676	.936	.785	.217
4	.642	.686	.837	.754	.120
5	.560	.648	.635	.641	.073
Mean	.599	.656	.761	.702	.087
SD	.061	.030	.124	.066	.090

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